UTS SOFTWARE CONSTRACTION
LAYANAN CRUD PRODUCT (PRODUCT SERVICE)

Oleh:

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Github: https://github.com/pandustrr/UTS-SoftwateConstraction

a. Deskripsi Singkat

Layanan ini adalah Product Service yang berfungsi untuk mengelola data produk dalam suatu sistem (misalnya sistem e-commerce atau inventaris). Layanan ini menyediakan operasi dasar CRUD (Create, Read, Update, Delete) melalui API yang dapat diakses secara publik.

b. Teknologi yang digunakan:
   - Framework: Laravel 11
   - Database: SQLite
   - Arsitektur: RESTful API

c. Penjelasan Endpoint API (CRUD)

Berikut adalah daftar endpoint yang tersedia dalam layanan ini:

Method	Endpoint	Deskripsi	Parameter
GET	/api/products	Mengambil semua data produk	-
POST	/api/products	Menambahkan produk baru	name, description, price, stock
GET	/api/products{id}	Mengambil detail produk berdasarkan ID	-
PUT/PATCH	/api/products{id}	Memperbarui data produk berdasarkan ID	name, description, price, stock
DELETE	/api/products{id}	Menghapus produk berdasarkan ID	-

d. Struktur JSON Request (Contoh POST/PUT)

```
json
{
    "name": "Kopi Susu Gula Aren",
    "description": "Kopi susu dengan gula aren pilihan",
    "price": 15000,
    "stock": 50
}
```

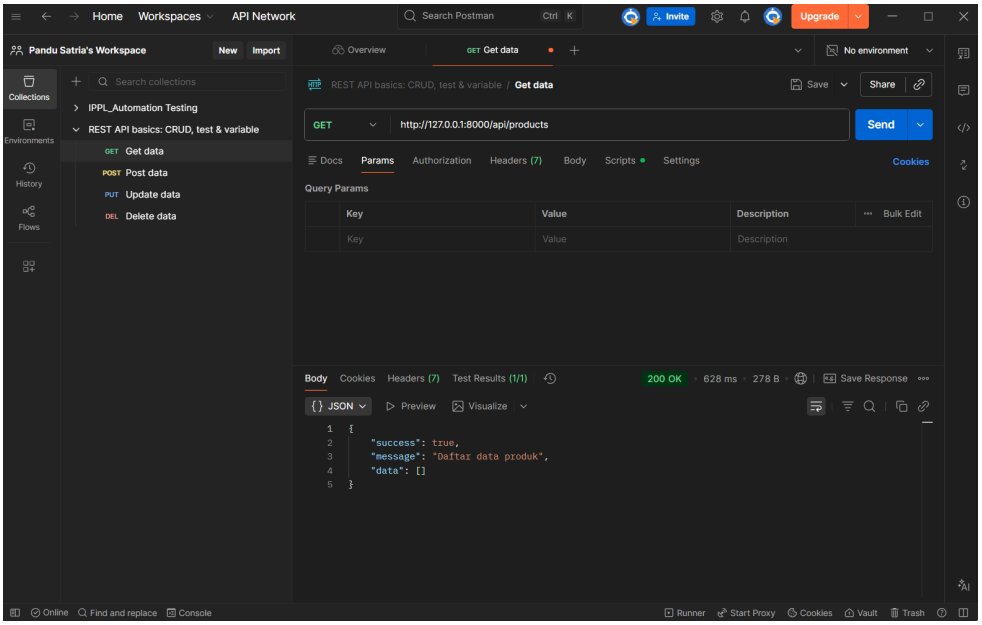
e. Pengujian Postman

1. Jalankan server lokal
   Buka terminal dan jalankan ‘php artisan serve” dan biasanya server akan berjalan di `http://127.0.0.1:8000`

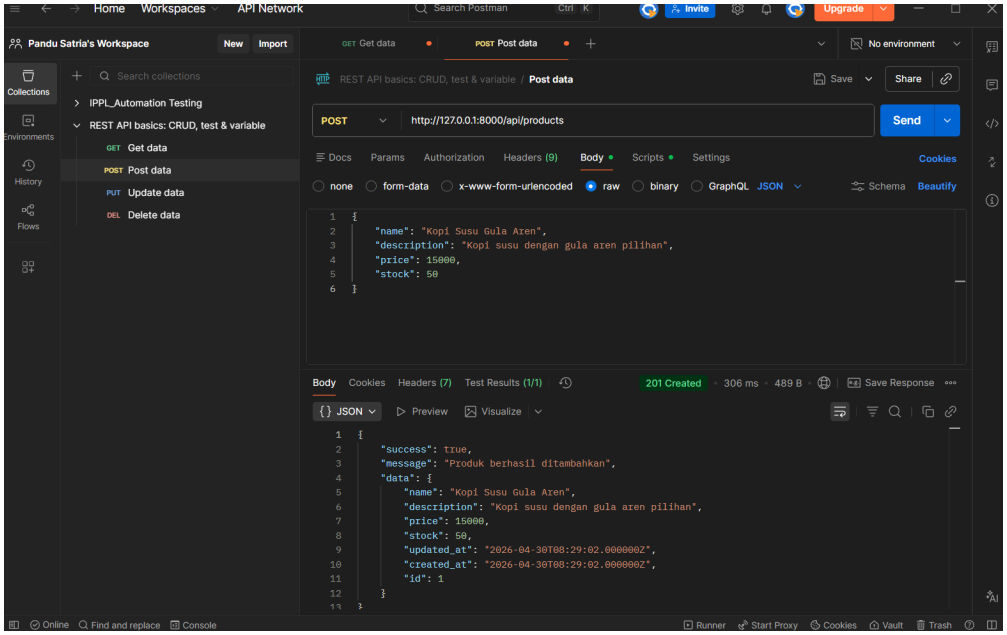
```
PS D:\KULIAH\SEMESTER 6\Sofware cons\UTS_SoftCons> php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].
  2026-04-30 14:50:58 / ..............................................
  2026-04-30 14:50:58 ................................................
  2026-04-30 14:51:04 /favicon.ico ...................................
```

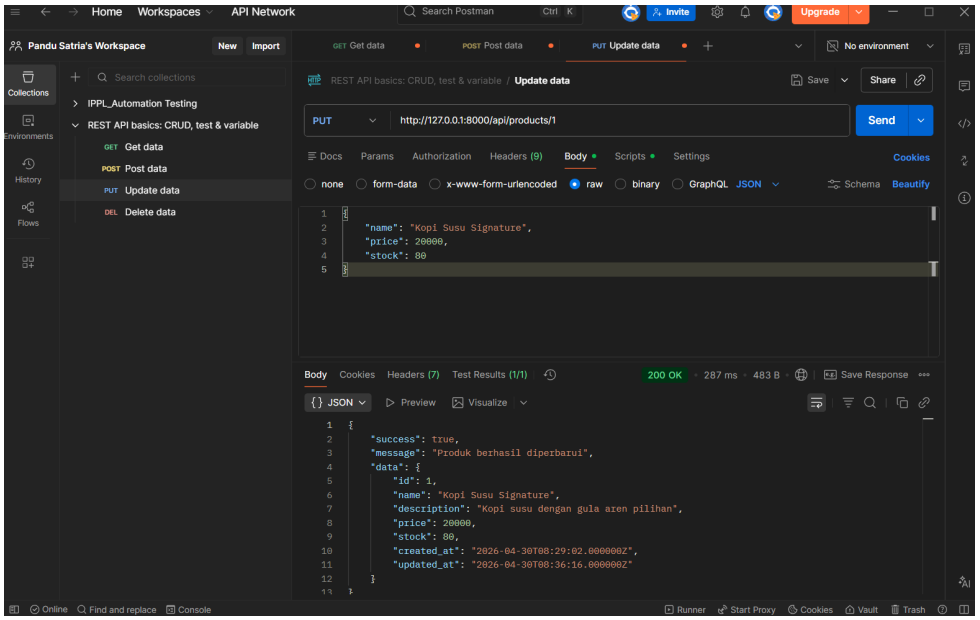
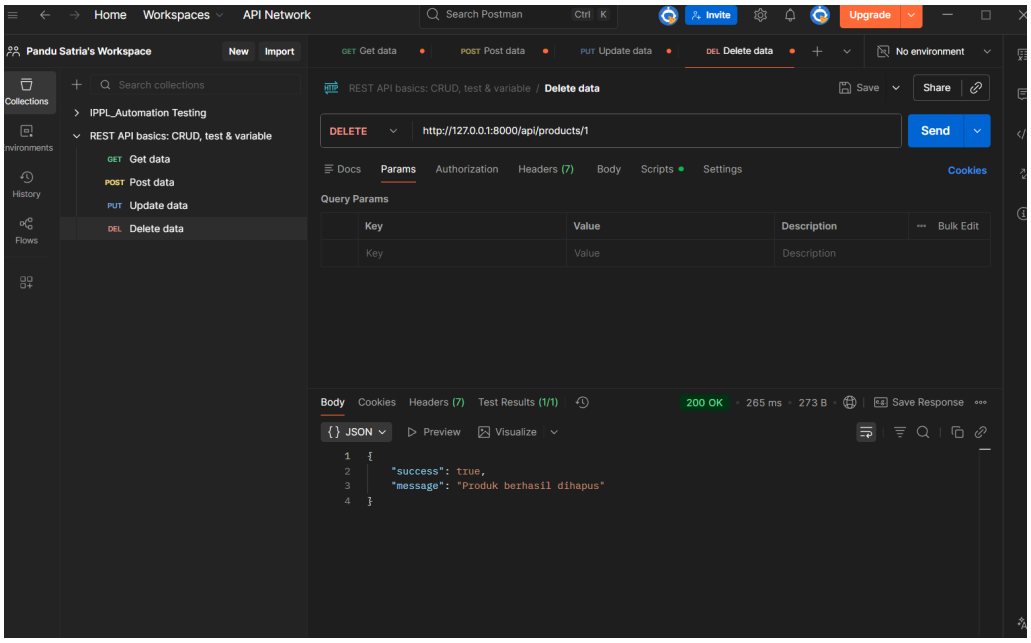
2. Get All Products
   - Method: GET
   - URL: `http://127.0.0.1:8000/api/products`
   - Klik Send



3. Create Product
    - Method: POST
    - URL: `http://127.0.0.1:8000/api/products`
    - Tab Body -> Pilih raw -> Pilih format JSON
    - Masukkan data JSON produk.
    - Klik Send



4. Update Product
    - Method: PUT
    - URL: `http://127.0.0.1:8000/api/products/1`
    - Tab Body -> Pilih raw -> Pilih format JSON
    - Masukkan data yang ingin diubah.
    - Klik Send



5. Delete Product
    - Method: DELETE
    - URL: `http://127.0.0.1:8000/api/products/{id}'
    - Klik: Send